Vipul Sapra

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EDUCATION

University of Cincinnati, Carl H. Lindner College of Business — **Cincinnati, OH**
Master of Science in Information Systems and Graduate Certificate in Data Science — **April 2022**
Cumulative GPA: 3.95 / 4.0

Pune University, Maharashtra Institute of Technology — **Pune, India**
Bachelor of Engineering in Petrochemical — **June 2017**

TECHNICAL SKILLS

Concepts: Cloud Data Engineering, Agentic AI & Prompt Orchestration, Generative AI & RAG, Autonomous Anomaly Detection, Lakehouse Architecture, Data SRE & Observability, MLOps & Model Monitoring, Distributed Computing, Dimensional Modeling (Star/Snowflake), CI/CD & Git-based Delivery, Data Governance & Lineage.

Programming Languages & Tools: AWS (S3, Redshift, Glue, EMR, Athena, IAM, KMS), Amazon SageMaker, Azure (ADF, Databricks, Synapse, Azure SQL), Snowflake, Apache Spark, PySpark, Python, SQL, Power BI (DAX), Tableau, Collibra, KIRO IDE, GitHub, Docker, Kubernetes.

Certifications: Data Science Graduate Certificate – University of Cincinnati

PROFESSIONAL EXPERIENCE

Charter Communications — Charlotte, NC

Senior Cloud Data & AI Platform Engineer — **May 2022 – Present**

- **Architected and deployed custom autonomous agentic workflows** (KIRO IDE, prompt orchestration, Copilot) to automate schema discovery, data profiling, regex classification, and code migration, reducing manual delivery turnaround by 40-50%.
- **Engineered autonomous AI anomaly detection frameworks** identifying schema variations, distribution drift, and ETL discrepancies with 95% precision using Python and PySpark, safeguarding downstream data lakehouse and BI integrity.
- **Built and deployed high-throughput distributed pipelines** and dimensional models (Star/Snowflake, SCD Type 1/2) using AWS (S3, Redshift), Azure, Snowflake, and Apache Spark to process hundreds of terabytes with 99.9% data availability SLAs.
- **Orchestrated advanced time-series and predictive machine learning models** (SARIMA, XGBoost, Scikit-Learn) for demand forecasting and customer behavior prediction, optimizing infrastructure capacity and improving service reliability by 20%.
- **Engineered automated testing frameworks for conversational AI** (SpectrumIQ) and integrated source-to-target reconciliation firewalls with Collibra data lineage, cutting manual QA turnaround by 50% and operational overhead by over 80%.

Data Scientist & Machine Learning Specialist — **May 2022 – Present**

- **Developed supervised and unsupervised machine learning algorithms** leveraging Python, Scikit-Learn, and XGBoost to extract high-value telemetry insights and drive proactive operational decision-making.
- **Designed scalable feature engineering pipelines** and integrated model outputs into enterprise data warehouses to power real-time predictive analytics and business intelligence.
- **Implemented rigorous model monitoring and validation strategies** ensuring high predictive accuracy, robustness, and compliance across production deployment cycles.

Business Intelligence & Decision Analytics Specialist — **May 2022 – Present**

- **Created executive Key Operating Metric (KOM) dashboards** and advanced semantic models using Power BI (DAX) and Tableau to translate complex transactional telemetry into actionable corporate strategy.
- **Optimized complex SQL queries and relational database schemas** to accelerate reporting performance and enhance data accessibility for enterprise stakeholders.
- **Collaborated with cross-functional teams to define data governance standards** and streamline reporting workflows, ensuring data consistency across business units.

Ameritas Insurance — Cincinnati, OH

Data Governance Intern — **September 2021 – April 2022**

- **Co-engineered the enterprise-wide Data Advocacy & Governance Program** across insurance portfolios, establishing foundational frameworks for data democratization and organizational accountability.
- **Architected a comprehensive Data Intelligence ecosystem in Collibra**, establishing a centralized Data Dictionary, automated Data Quality rules, and end-to-end **Data Lineage** to ensure regulatory compliance and transparency for financial and insurance assets.

Accenture — Pune, India

Test Engineering Analyst — **September 2017 – December 2020**

- **Orchestrated end-to-end Quality Assurance and system integration testing** for mission-critical client applications at **Allianz Insurance**, leveraging Agile SDLC methodologies to ensure rapid, zero-defect feature deployments.
- **Spearheaded the Health Insurance module QA strategy**, directing cross-functional efforts for regression testing, test data validation, and the seamless rollout of complex new business functionalities.
- **Drove operational excellence by automating core regression test suites**, significantly increasing engineering productivity and reducing manual testing overhead through scalable automation frameworks.